

Anomaly Detection Framework With Contrastive Learning and Multiview Augmentation for Time-Series Domain Generalization

Yeseul Lee^{ID}, Seunghwan Song^{ID}, Kwan-Yong Park^{ID}, Byoung-Mo Koo^{ID}, and Jun-Geol Baek^{ID}

Abstract—Time-series data are generated and collected in manufacturing processes. Time-series data have different distribution characteristics, with the underlying assumption being that all data originate from the same distribution. Time-series anomaly detection methods are trained on stationary data derived from the source domain. However, the performance of the trained model diminishes when confronted with a nonstationary distribution in the target domain. Consequently, there is a need for a model that can adapt to any distribution using only the source domain. In this study, we propose an anomaly detection framework with contrastive learning and multiview augmentation for time-series domain generalization. The proposed method uses multiview augmentation to learn three representations for time-series domain generalization. Moreover, the augmented samples use a temporal convolutional network (TCN) to extract these representations. Subsequently, the extracted representations are trained with a contrastive learning method inspired by simple triplet representation learning (SimTriplet). Therefore, the proposed method can attain the diversity and generalization performance of the source domain. The performance superiority of our proposed method is experimentally validated across various domain shift scenarios. Additionally, the effectiveness of the modules used in the proposed method is confirmed through qualitative analysis.

Index Terms—Contrastive learning, multiview augmentation, temporal convolutional network (TCN), time-series domain generalization.

I. INTRODUCTION

IN THE digital twin manufacturing process, the data are collected from several sensors. A large amount of normal data and a small amount of abnormal data are collected during the manufacturing process. The few abnormal data are related to the yield of the manufacturing process. Semisupervised anomaly detection aims to train the normality of a time-series representation under the assumption that most of the training data are normal [1]. However, these methods rely on

the distribution of the training data [2]. This is because most models assume that all the data are collected from the same distribution during training and testing. However, this strong assumption is not appropriate in the real world. In particular, distributions often change during manufacturing processes due to labor, machine changes, or maintenance. In this study, this situation is referred to as a domain shift. Since a domain shift causes performance degradation in existing models, appropriate measures are required [3].

Domain adaptation has been proposed to address domain shifts by transferring knowledge or information obtained from training data (i.e., the source domain) to a model based on test data (i.e., the target domain) [4], [5]. This process primarily involves reducing or adjusting the differences between the source and target domains, enabling the model to be trained in the source domain and adaptable to the target domain [6]. Despite its success in various tasks, domain adaptation still faces significant challenges. First, training in the target domain becomes unstable if the distribution is nonstationary. Domain adaptation aims to generalize from in-distribution (ID) data with stationarity. However, model performance degrades when the target domain distribution shifts to out-of-distribution (OOD) and becomes nonstationary, necessitating additional learning. Second, effective domain adaptation requires collecting samples from all possible distributions, which is inefficient and impractical [7]. Consequently, there is a need for models that can adapt to any distribution using only the source domains collected from the ID.

Domain generalization aims to maintain performance on unknown domains by training multiple independent source domains [8]. Domain adaptation focuses on minimizing the difference between the source and target domains; in contrast, domain generalization has been proposed as a technique to address the inherent limitations of domain adaptation. Domain generalization has been successful in a variety of vision tasks [9], [10] and natural language processing (NLP) tasks [11], [12], [13]. Representation learning is often used in domain generalization to extract a generalized representation for the source domain. Contrastive learning among representation learning has demonstrated effective performance for feature extraction [14], [15], [16]. Contrastive learning trains the same classes (i.e., positive pairs) to be closer in distance, while different classes (i.e., negative pairs) are further apart. Contrastive learning-based domain generalization studies typically aim to train domain-invariant

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and class-variant representations. Contrastive learning is an effective method for domain generalization, which is the problem of interest in this study. It affords several advantages over traditional methods, such as less dependence on labeled data, improved robustness to domain shifts, and utilization of effective data augmentation techniques for time-series data [15], [16]. Dou et al. [17] proposed model-agnostic learning of semantic features (MASF). This method can be used to extract domain-invariant and class-variant representations. In contrast, Kim et al. [16] overcame the instability of training due to negative pairs with self-supervised contrastive regularization (SelfReg), which uses only positive pair samples.

However, adapting contrastive learning-based domain generalization to time-series data presents several challenges. First, previous studies have focused on supervised learning or assumed fully annotated datasets [18]. In manufacturing processes, obtaining sufficient labeled data is time-consuming, making these approaches impractical. Second, many of these methods are designed for vision and NLP tasks, which utilize data augmentation techniques to diversify the source domain [19]. Contrastive learning-based domain generalization uses data augmentation to diversify the source domain used for training. Data augmentation is a technique used to increase the size of a practical dataset by diversifying the source domain [20]. Adding random noise, changing colors, and shapes have been used to augment the source domain [21], [22], [23]. However, because data augmentation is specialized for vision and NLP tasks, augmentation for time-series data is limited. Therefore, appropriate augmentation for time-series data is required.

Meanwhile, defining negative pairs in time-series data is difficult owing to the dependencies between time variables and inter-variables. Therefore, it is necessary to design a model that uses only positive pairs [24]. Liu et al. [25] proposed a simple triplet representation learning (SimTriplet) that uses only positive pairs. SimTriplet is a modified version of the exploring simple Siamese representation learning (SimSiam) [26] network structure. Compared to SimSiam, it has the advantage of learning multiple representations by adding features. SimTriplet uses the spatial property that geographically adjacent images are more likely to belong to the same class, achieved through two-view consistency learning, which minimizes the distance between two positive pairs. However, there are limitations when applied to time-series data, as they lack distinct classes and spatial features.

In this study, we proposed an anomaly detection framework with contrastive learning and multiview augmentation for time-series domain generalization. The proposed method aims to detect time-series anomalies in a domain shift situation by extracting a generalized representation. First, the proposed method uses multiview augmentation for time-series data. Multiview augmentation learns three representations for time-series domain generalization: 1) domain-invariant representation; 2) robust representation; and 3) task-adaptive representation. These representations have the advantage of providing source domain diversity and generalization

performance. Then, the augmented sample is processed using a temporal convolutional network (TCN). A TCN is a convolutional neural network (CNN)-based model that can be used for time-series modeling and has the advantage of fast computation speed [27]. Finally, the extracted representation is trained using a contrastive learning method inspired by SimTriplet [25].

The key contributions of this study are as follows:

- 1) multiview augmentation improves domain generalization by generating multiple representations of time-series data;
- 2) the TCN and SimTriplet structures make training suitable for time-series data; and
- 3) domain generalization experiments on various time-series datasets demonstrate the robustness of the proposed method.

The remainder of this study is organized as follows. Section II provides a detailed description of the proposed method and its structure. Section III presents an evaluation of three different time-series datasets using the proposed method to verify its performance. Finally, Section IV presents the conclusions and future works of this study.

II. PROPOSED METHOD

In this section, we propose an anomaly detection framework with contrastive learning and multiview augmentation for time-series domain generalization. The proposed method draws inspiration from SimTriplet [25] but undergoes a model redesign to better suit the requirements of time-series anomaly detection. Initially, multiview augmentation is employed to enhance the diversity of time-series data in the source domain. Subsequently, the TCN encoder is utilized to extract representations for each augmented time-series dataset. The overall structure of the proposed method is depicted in Fig. 1.

A. Multiview Augmentation

Effective time-series domain generalization necessitates achieving three primary objectives [14]. First, the consideration of domain-invariant representations between the source domains allows for the learning of generalized representations across domain variations without being constrained to a specific domain. Second, the model must exhibit robustness against external environments, ensuring stable training under diverse conditions by accounting for noise. Lastly, it must possess task-adaptive capabilities to effectively perform tasks in the target domain, making the model applicable to real-world scenarios. In this study, we utilized three augmentations to perform domain generalization in time-series anomaly detection. First, we extracted general features from multiple tasks in the source domain; this helps achieve performance in the target domain. It also enables the extraction of representations that are robust to real-world, noisy time-series anomalies.

To attain these objectives, multiview augmentation is employed, incorporating adversarial attacks and jitter-based data augmentation. Adversarial attacks for data augmentation are implemented using the fast gradient sign method (FGSM)

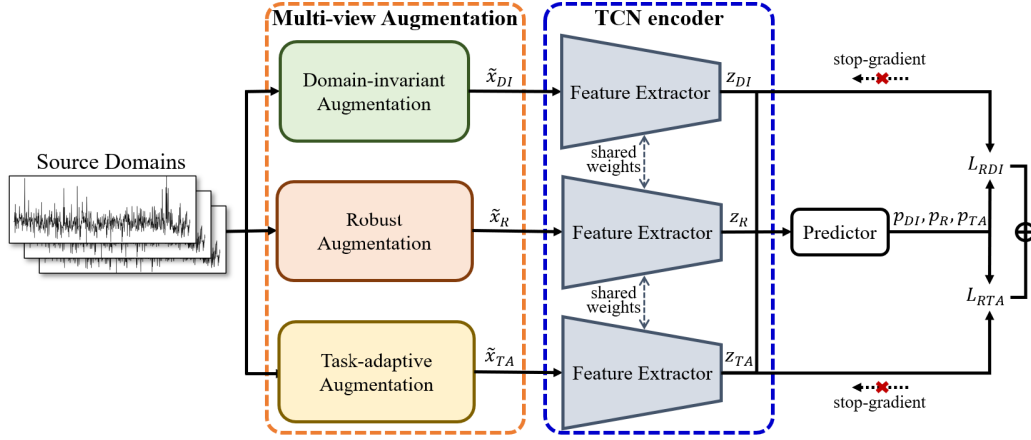


Fig. 1. Contrastive learning with multiview augmentation for time-series domain generalization framework.

[28] in this study. The FGSM-based data augmentation is expressed as follows:

$$x_{\text{adversarial}} = x + \epsilon_{\text{adversarial}} \cdot \text{sign}(\nabla_x L(\theta, x, y)) \quad (1)$$

where x and y denote the train data and labels, respectively, θ is the weight of the network, and $L(\cdot)$ is the loss function. The process for creating an adversarial sample using the FGSM is as follows:

Step 1: Calculate the gradient $\nabla_x L(\theta, x, y)$ for the loss function on the training data.

Step 2: The gradient $\nabla_x L(\theta, x, y)$ is updated in a direction opposite to the label y .

Step 3: The noise $\text{sign}(\nabla_x L(\theta, x, y))$ that can cause misclassification during the update process is defined.

Step 4: The noise $\text{sign}(\nabla_x L(\theta, x, y))$ is adjusted by a hyperparameter $\epsilon_{\text{adversarial}}$, which is added to the original data x .

FGSM-based data augmentation was used for source domain-invariant and task-adaptive representations. This is achieved by updating the noise values obtained from each model in the existing FGSM equation. By contrast, jitter-based augmentation is used for external robustness.

1) Domain-Invariant Representation: If the model is trained on common and general patterns across the source domains, it can achieve performance in the target domain. This implies that training the model to distinguish between source domains prevents overreliance on a particular domain. In this study, we utilized FGSM-based data augmentation. FGSM trains the source domain-invariant representation by adding noise to the original data [28]. This process increases the uncertainty of the model because it generates an adversarial sample that confuses the model, leading to misclassification. In this study, we employed long short-term memory (LSTM) [29] as a classification model to consider the temporal pattern, and cross-entropy served as the loss function. Consequently, the creation of an adversarial sample for a domain-invariant representation is shown in the following equation:

$$\tilde{x}_{DI} = x + \epsilon_{DI} \cdot \text{sign}(\nabla_x L(\theta^S, D^S, y^S)) \quad (2)$$

where x is the original data and D^S and y^S are the data and labels in the source domain, respectively. In addition, θ^S is the source domain network weights and $\nabla_x L(\theta^S, D^S, y^S)$

denotes the gradient of the loss function. The noise $\text{sign}(\nabla_x L(\theta^S, D^S, y^S))$ is adjusted by the hyperparameter ϵ_{DI} . Consequently, $\tilde{x}_{DI}$ trains the source domain-invariant representation by adding noise, which can lead to misclassification.

2) Robust Representation: In the real world, a model that can reliably perform in various environments is required. In particular, considerable noise is generated during the manufacturing process, making it crucial to ensure the robustness of generalized performance under diverse conditions and environments. In this study, we employed jitter-based data augmentation, as it has been demonstrated to effectively handle real-world noise in several studies [30]. The creation of a sample for a robust representation is shown in the following equation:

$$\tilde{x}_R = x + \epsilon_R \cdot \sigma \quad (3)$$

where x is the original data and σ is the standard deviation for each variable. The standard deviation σ is adjusted by the hyperparameter ϵ_R . Consequently, $\tilde{x}_R$ trains the robust representation to deal with real-world noise and variation.

3) Task-Adaptive Representation: Time-series domain generalization must be task-adaptive to the target domain, signifying that the proposed model can be applied in new environments. For this purpose, we utilized inverse FGSM (InvFGSM) [28]. InvFGSM enhances task-adaptive performance by subtracting the noise of the model from the original data. In this process, InvFGSM aids in achieving better performance by adjusting the inputs with minimal loss [31]. Therefore, InvFGSM-based data augmentation can enhance anomaly detection performance in the target domain. In this study, we employed a temporal convolutional autoencoder (TCN-AE) [32] as an anomaly detection model, with the mean squared error (mse) serving as the loss function. The creation of an adversarial sample for a task-adaptive representation is as shown in the following equation:

$$\tilde{x}_{TA} = x - \epsilon_{TA} \cdot \text{sign}(\nabla_x L(\theta_i^S, D_i^S)), \quad i = 1, \dots, M \quad (4)$$

where x is the original data, and D_i^S is the i th source domain. The total number of source domains is M . In addition, θ_i^S is the i th source domain network weights and $\nabla_x L(\theta_i^S, D_i^S)$ denotes the gradient of the i th loss function. The i th noise $\text{sign}(\nabla_x L(\theta_i^S, D_i^S))$ is adjusted by the hyperparameter ϵ_{TA} .

Consequently, $\tilde{x}_{TA}$ trained the task-adaptive representation to allow the model to achieve more robust results.

B. TCN Encoder for Feature Extraction and Predictor

In this study, we employed a TCN encoder for feature extraction from the augmented samples. TCN is a CNN-based model that can be used in sequence modeling and uses dilated convolution [27]. Dilated convolution has the advantage of increasing the receptive field to obtain long time step information without the need for a convolutional layer. TCN is suitable for time-series modeling and offers the advantage of faster computation compared to a recurrent neural network (RNN)-based models [27]. The proposed method utilized three parallel TCN encoders with shared weights, extracting features from the multiview augmented samples. In this study, we utilized ten dilated convolution layers in the TCN encoder, enabling the extraction of generalized representations from long-term timesteps. As a result, TCN encoders could process each sequential input and capture the relationship between the augmented samples.

The predictor and stop-gradient are devices to prevent mode collapse that occurs when SimTriplet performs contrastive learning using only positive pairs. The predictor utilizes a multilayer perceptron structure with two linear layers, which is the same as the SimTriplet structure.

C. Loss Function

In domain generalization, training and testing are performed using the source and target domains, respectively. In this study, we trained an encoder that can extract the representation with the loss function negative cosine similarity (NCS) loss using the source domain data in the training process. In the test process, we defined the Euclidean distance between the target domain representation and the source domain representation obtained by the encoder as the anomaly score and utilized it for anomaly detection.

Our model for contrastive-learning-based time-series domain generalization relies on positive pairs exclusively, drawing inspiration from the SimTriplet [25]. The suitability of two-view consistency learning in SimTriplet for learning the relationship between augmented samples motivated this choice. Consequently, each output of the TCN encoder serves as an asymmetrical predictor, with a stop gradient applied. The NCS [33] is employed to calculate dissimilarity between the outputs, reducing differences between the augmented samples through opposite characteristics. The NCS is shown in the following equation:

$$\text{NCS}(p, q) = -\frac{p}{\|p\|_2} \cdot \frac{q}{\|q\|_2}. \quad (5)$$

In (5), p and q denote two vectors. Further, $\|p\|_2$ and $\|q\|_2$ denote the Euclidean distances between the two vectors. NCS makes the two vectors more similar and ranges from -1 to 1 .

In this study, $\tilde{x}_{DI}$, $\tilde{x}_R$, and $\tilde{x}_{TA}$ were created using multiview augmentation. These representations are processed using the TCN encoder to obtain z_{DI} , z_R , and z_{TA} . These features were passed on to the predictor and expressed

as p_{DI} , p_R , and p_{TA} . The proposed method involves two loss functions for time-series domain generalization: 1) minimizing the dissimilarity between the source domain-invariant representation and the robust representation to ensure that the learned features are stable and generalizable across different domains and 2) minimizing the dissimilarity between the task-adaptive representation for the target domain and the robust representation to enhance the model's ability to adapt to new tasks and improve anomaly detection performance in the target domain. Each loss function is expressed as follows:

$$L_{RDI} = \frac{1}{2} [\text{NCS}(z_{DI}, p_R) + \text{NCS}(z_R, p_{DI})] \quad (6)$$

$$L_{RTA} = \frac{1}{2} [\text{NCS}(z_{TA}, p_R) + \text{NCS}(z_R, p_{TA})]. \quad (7)$$

In (6), L_{RDI} maximizes the similarity between the source domain-invariant representation and the robust representation, promoting a robust domain-invariant to facilitate learning and avoid misclassification among between different source domains. Conversely, L_{RTA} in (7) maximizes the similarity between the task-adaptive representation for the target domain and the robust representation, fostering robust task adaptation to enhance the extraction of representations for improved anomaly detection performance in the target domain. The final loss function of the proposed method is shown in the following equation:

$$L_{\text{Total}} = L_{RDI} + L_{RTA}. \quad (8)$$

The final loss function of the proposed method helps TCN encoders with shared weights in embedding the representations of different augmentations. Meanwhile, the stop-gradient is accomplished by stopping or not propagating a particular gradient during backpropagation [26]. Therefore, the proposed method can be trained to avoid mode collapse through the final loss function and to stop the gradient.

Consequently, the output of the proposed method is the latent vector calculated by the TCN. Training of the proposed method yields a feature extractor that is task-adaptive and capable of extracting domain-invariant representations.

III. EXPERIMENT

A. Datasets

The aim of the proposed method is to train a representation for time-series data in the source domain, enabling effective time-series anomaly detection, even in a domain shift situation in the target domain. Therefore, two datasets were selected to evaluate the proposed method.

1) *SKAB Dataset*: The Skoltech Anomaly Benchmark (SKAB) dataset was collected during the simulation of a water circulation system [34]. It comprises Valve1 at the outlet and Valve2 at the inlet of the water circulation pump. Each subset of Valve1 and Valve2 signifies the time order in which they were collected. There were 14 subsets of Valve1 and four subsets of Valve2. The data consist of eight variables (accelerometer1, accelerometer2, temperature, thermocouple, current, pressure, voltage, and rate) along with a time step. The SKAB dataset contains the following anomaly types: abrupt change, contextual anomalies, collective anomalies, and

TABLE I
TASK INFORMATION FOR TIME, MACHINE, AND
OPERATION DOMAIN SHIFT SCENARIO

Scenario	Task	Source domain	# of source domain	Target domain	Dataset
Time domain shift	T1	Valve1_00~06	7	Valve1_07	SKAB
	T2	Valve1_00~07	8	Valve1_08	
	T3	Valve1_00~08	9	Valve1_09	
	T4	Valve1_00~09	10	Valve1_10	
	T5	Valve1_00~10	11	Valve1_11	
	T6	Valve1_00~11	12	Valve1_12	
	T7	Valve1_00~12	13	Valve1_13	
Machine domain shift	M1	Machine02, 03	2	Machine01	CNC milling machine
	M2	Machine01, 03	2	Machine02	
	M3	Machine01, 02	2	Machine03	
Operation domain shift	O1	w/o Operation02	14	Operation02	
	O2	w/o Operation04	14	Operation04	
	O3	w/o Operation07	14	Operation07	
	O4	w/o Operation10	14	Operation10	

gradual change. Additionally, both normal and abnormal data are labeled, with a 35% anomaly ratio.

2) *CNC Milling Machine Dataset*: The computer numerical control (CNC) milling machine dataset was collected during the processing of a piece of aluminum [34]. It includes three machine subsets and 15 operation subsets, reflecting the variability in the physical flow of production during the manufacturing process. The data consist of three variables (x -axis, y -axis, and z -axis vibrations) along with a time step. The CNC milling machine dataset contains the following anomaly types: tool wear, spindle malfunction, unexpected vibrations, temperature anomalies, and misalignment. Normal and abnormal data are labeled, with a 4.12% anomaly ratio.

B. Problem Definition

The proposed method aims for good performance in the target domain by training in the source domain. Therefore, three domain shift scenarios were considered using the SKAB and CNC milling machine datasets. For the time domain shift scenario, the SKAB dataset is considered because it contains valves collected at different times in the same environment. For the machine and operation domain shift scenarios, the CNC milling machine dataset is considered as it contains information about different machines or operations collected simultaneously. The task information for each domain shift scenario is presented in Table I.

The time domain shift assesses the performance of the proposed method as the temporal distribution undergoes changes. Valve1 was considered because it contained sufficient subsets of the SKAB dataset. There are 14 subsets of Valve1 ranging from Valve1_00 to Valve1_13, where “ t ” in each Valve1_ t signifies the temporal order. Tasks for evaluating the time domain shift scenario were categorized from T1 to T7, with the number of source domains increasing by one for each task to align with the temporal order. The target domain consists of a subset immediately after each task.

The machine domain shift assesses the performance of the proposed method as the distribution of machines

changes. Three different machines Machine01, Machine02, and Machine03 of a CNC milling machine were considered for this purpose. Three tasks (M1, M2, and M3) were established to verify the machine domain shift, with each task’s source domain comprising two machines and the target domain consisting of the remaining machine not used in the source domain.

The operation domain shift evaluates the proposed method’s performance when the operating conditions of the machine undergo changes. Fifteen different operations (Operation01 to Operation15) from the CNC milling machine dataset were considered for this purpose. Four tasks (O1, O2, O3, and O4) were employed to verify operational domain shifts. In each task, the source domain included the remaining 14 operations, excluding one specific operation. In this study, Operation02, Operation04, Operation07, and Operation10 were chosen as specific operations because only four out of the 15 contain both normal and abnormal data.

C. Experiment Details

Time-series domain generalization requires testing of distributions that are not used in training. Therefore, M source domains $D^S = [D_1^S, \dots, D_M^S]$ with only normal data used in the training process. The testing process used the target domain D^T that was not part of the training, containing both normal and abnormal data. This setup allowed the evaluation of the performance of the target domain for a model trained in the source domain.

The proposed method aims to achieve generalized performance in time-series anomaly detection. Consequently, various time-series anomaly detection models were employed for comparison: deep support vector data description (Deep SVDD) [36], multiscale convolutional recurrent encoder–decoder (MSCRED) [37], unsupervised anomaly detection (USAD) [38], deep transformer networks for anomaly detection (TranAD) [39], and SimSiam [26]. All algorithms were trained for 50 epochs with a window size of 60 and a batch size of 64, and each experiment was repeated five times. The experiments were implemented using the PyTorch software package in the Python3 language. Training procedures utilized an Intel¹ Core² i9-12900K CPU at 3.60 GHz, 64 GB of RAM, and an NVIDIA GeForce RTX 3090Ti graphics card with 32 GB of G6X memory.

D. Evaluation Metrics

Two metrics were employed to evaluate the performance of time-series domain generalization. First, the area under the receiver operating characteristic (AUROC) curve assessed the anomaly detection performance. A higher AUROC value indicates effective discrimination between normal and abnormal data. Second, to evaluate generalization performance in domain shift situations, a new metric, the GP-Score, was defined as follows:

$$\text{GP-Score} = \frac{\text{AUROC of source to target domain}}{\text{AUROC of source to source domain}}. \quad (9)$$

¹Registered trademark.

²Trademarked.

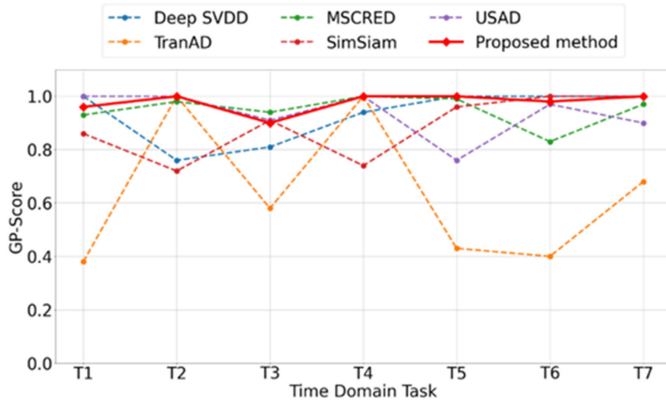


Fig. 2. GP-Score in time domain shift.

The GP-Score is calculated as the ratio of the AUROC for different types of test data when the model is trained on normal data from the source domain. In (9), the denominator represents the AUROC when the model is trained on normal data from the source domain and tested on normal and abnormal data from the source domain. The numerator represents the AUROC when the model is trained on normal data from the source domain and tested on normal and abnormal data from the target domain. Essentially, the GP-Score quantifies the performance change as the distribution shifts. Any GP-Score greater than 1 is scaled to 1, enabling the observation of quantitative performance changes in domain shift situations.

E. Experiment Results

1) *Time Domain Shift Result:* To assess the performance of the proposed method under changes in the temporal distribution, seven tasks (T1–T7) were considered using the SKAB dataset. The source domains were utilized for training in chronological order, while the target domain is a subset immediately after each task. Table II and Fig. 2 present the AUROC and GP-Score results for the time domain shifts, respectively.

The experimental findings indicate that the proposed method outperforms the comparison models in four out of seven tasks. For the remaining three tasks, the proposed method exhibits less performance degradation than the comparison models. In contrast, anomaly detection models based on adversarial learning, such as USAD and TranAD, exhibit limitations in their ability to learn representations from source domains. TranAD performs poorly due to overfitting to the source domains caused by the transformer structure used in the model. Additionally, deep SVDD performs inadequately as it involves converting multiple source domains into a single center. The GP-Score suggests that the proposed method maintains stable performance under time domain shifts, indicating its ability to sustain generalized performance even when the temporal distribution changes.

2) *Machine Domain Shift Result:* To evaluate the performance of the proposed method under changes in the machine distribution, three tasks (M1, M2, and M3) were considered using the CNC milling machine dataset. Each task's source domain comprised two machines, and the target domain

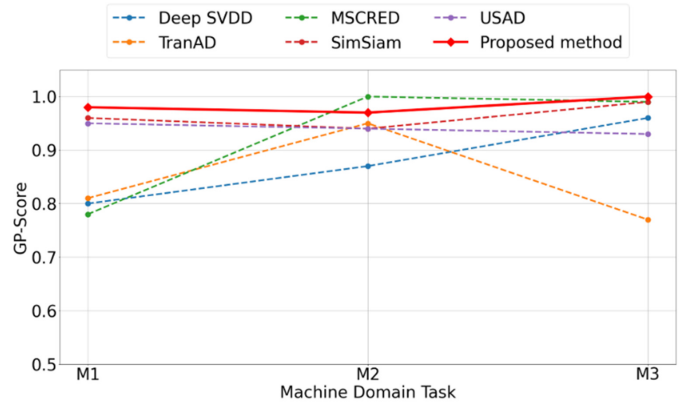


Fig. 3. GP-Score in machine domain shift.

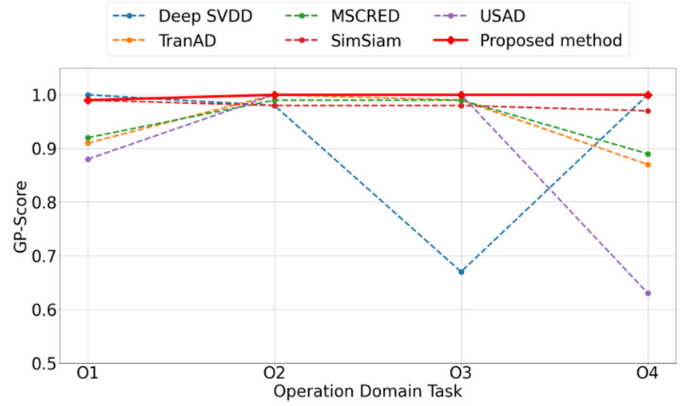


Fig. 4. GP-Score in operation domain shift.

included the remaining unused machines in the source domain. Table III and Fig. 3 present the AUROC and GP-Score results, respectively, for the machine domain shifts.

Experimental results reveal that the proposed method performs the best in two out of the three tasks. The AUROC performance for task M2 was suboptimal; however, the GP-Score results remained stable compared to those of other models. The comparison models, except for Deep SVDD, demonstrated effective performance under machine domain shifts. This is attributed to the CNC milling machine dataset not inherently being prone to machine domain shifts. Consequently, the results indicate that the proposed method extracts general representations from multiple source domains and adapts to variations in the machine distribution, suggesting that stable performance can be achieved with a small number of source domains.

3) *Operation Domain Shift Result:* To assess the performance of the proposed method under changes in machine operating conditions, four tasks (O1, O2, O3, and O4) were considered using the CNC milling machine dataset. The source domain for each task included all 14 operations, except for one specific operation, while the target domain comprised the remaining unused operations in the source domain. Table IV and Fig. 4 present the AUROC and GP-Score results, respectively, for the operation domain shifts.

The experimental results showed that the proposed method performed the best in all tasks. Because the experiment on the operational domain shifts trains 14 source domains, most of the comparison models also performed well. However,

TABLE II
AUROC OF TIME-SERIES DOMAIN GENERALIZATION IN TIME DOMAIN SHIFT (SKAB DATASET)

Method	Task							Avg. (Std.)
	T1	T2	T3	T4	T5	T6	T7	Average
Deep SVDD	0.67(0.04)	0.50(0.10)	0.52(0.03)	0.66(0.07)	0.68(0.03)	0.61(0.05)	0.67(0.05)	0.62(0.05)
MSCRED	0.79(0.08)	0.89(0.03)	0.83(0.06)	0.99(0.01)	0.89(0.05)	0.65(0.14)	0.92(0.06)	0.85(0.08)
USAD	0.84(0.06)	0.93(0.02)	0.67(0.09)	0.87(0.04)	0.61(0.15)	0.76(0.18)	0.72(0.04)	0.81(0.08)
TranAD	0.27(0.18)	0.79(0.18)	0.40(0.29)	0.87(0.13)	0.29(0.08)	0.24(0.16)	0.46(0.12)	0.47(0.16)
SimSiam	0.70(0.04)	0.71(0.03)	0.62(0.03)	0.73(0.03)	0.75(0.02)	0.74(0.03)	0.76(0.03)	0.72(0.03)
Proposed method	0.82(0.06)	1.00(0.00)	0.72(0.08)	0.99(0.01)	0.85(0.06)	0.88(0.06)	0.97(0.03)	0.88(0.04)

TABLE III
AUROC OF TIME-SERIES DOMAIN GENERALIZATION IN MACHINE DOMAIN SHIFT (CNC MILLING MACHINE DATASET)

Method	Task			Avg. (Std.)
	M1	M2	M3	Average
Deep SVDD	0.61(0.07)	0.50(0.02)	0.87(0.07)	0.66(0.05)
MSCRED	0.74(0.22)	0.99(0.00)	0.99(0.00)	0.91(0.09)
USAD	0.93(0.05)	0.94(0.01)	0.92(0.05)	0.93(0.04)
TranAD	0.80(0.20)	0.95(0.02)	0.76(0.23)	0.84(0.15)
SimSiam	0.75(0.03)	0.80(0.01)	0.99(0.00)	0.85(0.01)
Proposed method	0.95(0.02)	0.92(0.02)	1.00(0.00)	0.96(0.01)

TABLE IV
AUROC OF TIME-SERIES DOMAIN GENERALIZATION IN OPERATION DOMAIN SHIFT (CNC MILLING MACHINE DATASET)

Method	Task				Avg. (Std.)
	O1	O2	O3	O4	Average
Deep SVDD	0.50(0.00)	0.51(0.01)	0.52(0.01)	0.50(0.00)	0.51(0.01)
MSCRED	0.99(0.01)	0.99(0.00)	0.99(0.01)	0.93(0.07)	0.97(0.02)
USAD	0.84(0.01)	1.00(0.00)	1.00(0.00)	0.28(0.01)	0.78(0.06)
TranAD	0.92(0.07)	1.00(0.00)	0.97(0.03)	0.87(0.12)	0.94(0.06)
SimSiam	0.99(0.01)	0.98(0.01)	0.98(0.01)	0.95(0.02)	0.98(0.01)
Proposed method	0.99(0.01)	1.00(0.00)	1.00(0.00)	0.98(0.01)	0.99(0.00)

the performance of most models tended to drop sharply at O4. This is because Operation10, which is used as the target domain in O4, has a greater variation in the operation distribution than the source domains. In contrast, the GP-Score shows that the proposed method performs consistently and well in all tasks. This demonstrates that the proposed method maintained its generalized performance when the machine operation distribution changed.

F. Ablation Study

We performed an ablation study involving a qualitative analysis of the proposed method to validate the effectiveness and necessity of the employed modules. Multiview augmentation achieves source domain-invariant, robust, and task-adaptive representations in the target domain. Fig. 5 illustrates the individual impact of multiview augmentation and the sensitivity of each augmented sample to hyperparameters ϵ_{DI} , ϵ_R , and ϵ_{TA} .

Fig. 5(a) shows the AUROC performance as a result of the individual impacts of multiview augmentations $\tilde{x}_{DI}$ and $\tilde{x}_{TA}$. The bar graph shows the mean, and the line graph shows the

standard deviation. The experiments were performed on the SKAB dataset in a time domain shift scenario. Elucidating the individual effects of multiview augmentation is difficult because the framework used in this study consists of shared weights of the three TCN encoders. Therefore, we replaced the augmentation with jitter-based augmentation to verify its effectiveness. Experiments showed that the best performance is achieved when the multiview augmentation technique is fully utilized. In addition, $\tilde{x}_{DI}$ had the greatest impact on multiview augmentation.

Fig. 5(b) shows the sensitivity of each augmented sample to the hyperparameters ϵ_{DI} , ϵ_R , and ϵ_{TA} of the multiview augmentation. The box plot represents the AUROC of the seven tasks in the SKAB dataset. The experimental results show that $\tilde{x}_{DI}$ and $\tilde{x}_{TA}$ are hardly affected by the hyperparameters, while $\tilde{x}_R$ is considerably affected. Consequently, multiview augmentation is relatively robust to hyperparameters. This can lead to stable performance in time-series domain generalization. The proposed method used fixed hyperparameters regardless of the dataset: $\epsilon_{DI} = 0.001$, $\epsilon_R = 0.0005$, and $\epsilon_{TA} = 0.005$. The hyperparameter values were determined through extensive experimentation

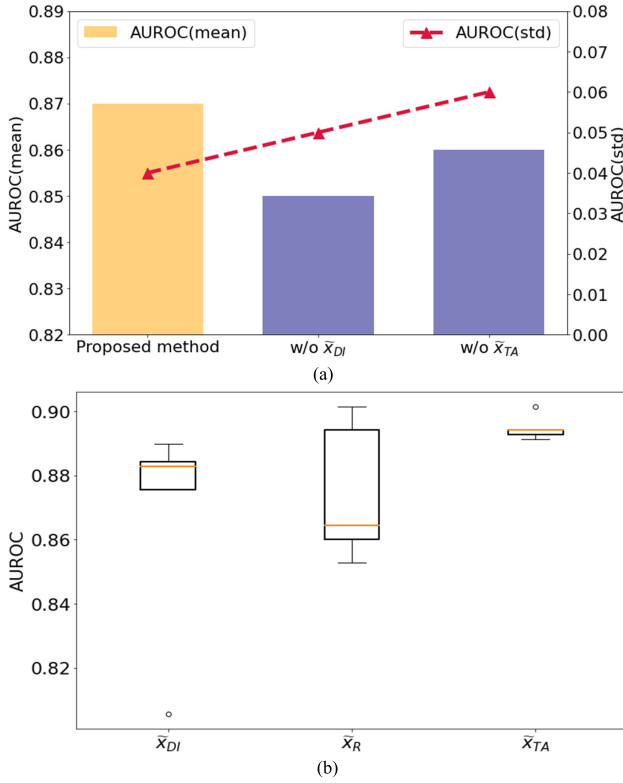


Fig. 5. Ablation study of multiview augmentation. (a) Comparison of the impacts of multiview augmentation. (b) Hyperparameter sensitivity in multiview augmentation.

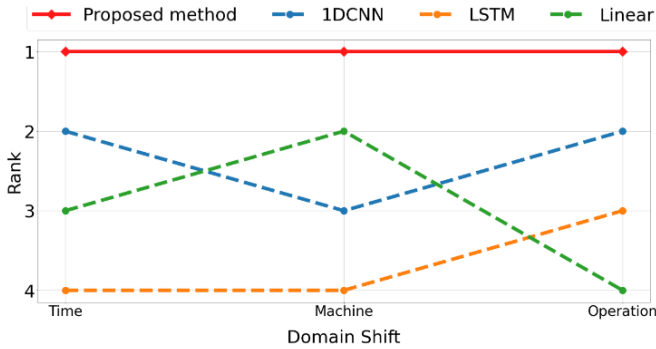


Fig. 6. Ablation study for feature extraction.

to balance model performance in different domains and tasks.

In this study, we utilized a TCN encoder for feature extraction for the augmented samples. Comparison with other encoder methods demonstrated the effectiveness of the TCN encoder for time-series modeling. Three time-series encoders were used for comparison: 1-D CNN, LSTM, and linear. Fig. 6 presents the ranking results for the AUROC performance upon changing the encoder in the proposed method.

For the domain shift scenario, we performed investigations by changing the encoder used for feature extraction. We performed five experiments with each of the encoders and ranked the results. As shown in Fig. 6, the best results corresponded to the use of the TCN encoder on all datasets. This shows that the TCN encoder can effectively extract features from time-series data.

IV. CONCLUSION

This study devised a novel anomaly detection framework that utilizes multiview augmentation to enhance time-series domain generalization. This structure leverages multiview augmentation to learn three features for time-series domain generalization. Subsequently, the time-series features are extracted using a TCN encoder. Finally, the extracted representations are trained using a contrastive learning method inspired by SimTriplet. Experiments were conducted in three domain shift scenarios to evaluate the performance of the proposed method; the results demonstrated its effectiveness in learning generalized features from various source domains to perform reliably within the target domain. Moreover, the proposed method exhibited high generalization performance compared to existing methods. Finally, qualitative analysis was performed to verify the effectiveness of the modules used in the proposed method.

Despite the promising results, a few limitations must be acknowledged. First, the proposed method heavily relies on the quality and diversity of the source domain data. If the source domains are not adequately representative of potential target domains, the model's generalization performance may suffer. Second, while the method shows robustness in simulated domain shifts, its performance in real-world applications, where domain shifts can be more unpredictable and complex, requires further validation. Future studies should focus on addressing these limitations by exploring more efficient algorithms, incorporating additional real-world datasets for validation, and enhancing the adaptability of the model to unforeseen domain shifts.

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